MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2014
Higher 1

CANDIDATE NAME	

CIVICS GROUP	1	3					INDEX NUMBER		

H1 BIOLOGY 8875/02

Paper 2

18 September 2014
2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination,
1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
   - Section A
   - Section B

The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 10
2	/ 10
3	/ 11
4	/ 9
Section B	/ 20
Total	**/60**

This paper consists of **12** printed pages.

[Turn over]

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows the tertiary structure of a molecule Ribonuclease, an enzyme that degrades RNA. The letter C represents cysteine, a sulphur containing amino acid. The numbers represent positions along the polypeptide chain. Number 40, for example, is the 40th amino acid.

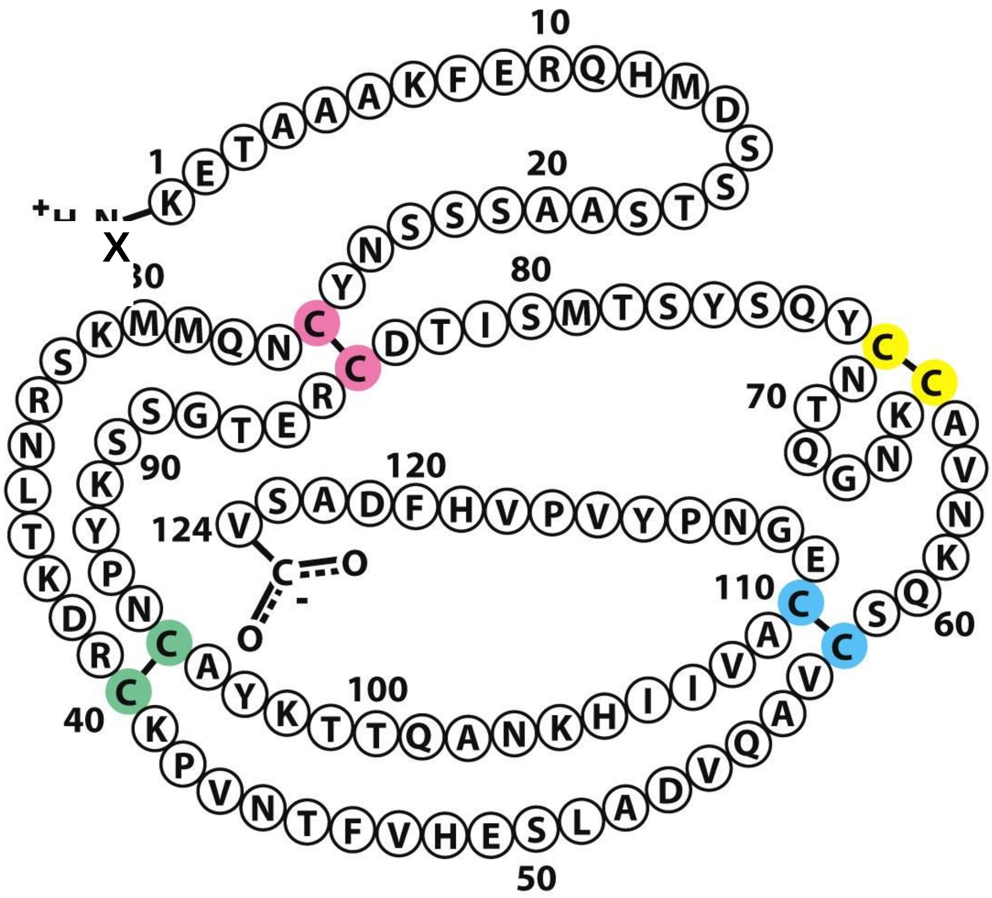

Fig. 1.1

(a) With reference to Fig. 1.1,

(i) Name the chemical group represented by the letter **X**, attached to the first amino acid. [1]

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(ii) Explain why all molecules of Ribonucleases have the same tertiary structure. [1]

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In an investigation on the importance of the tertiary structure of enzymes, Ribonuclease was treated with a mixture of mercaptoethanol and urea. After treatment, the enzyme activity was measured. The results are shown in Fig. 1.2.

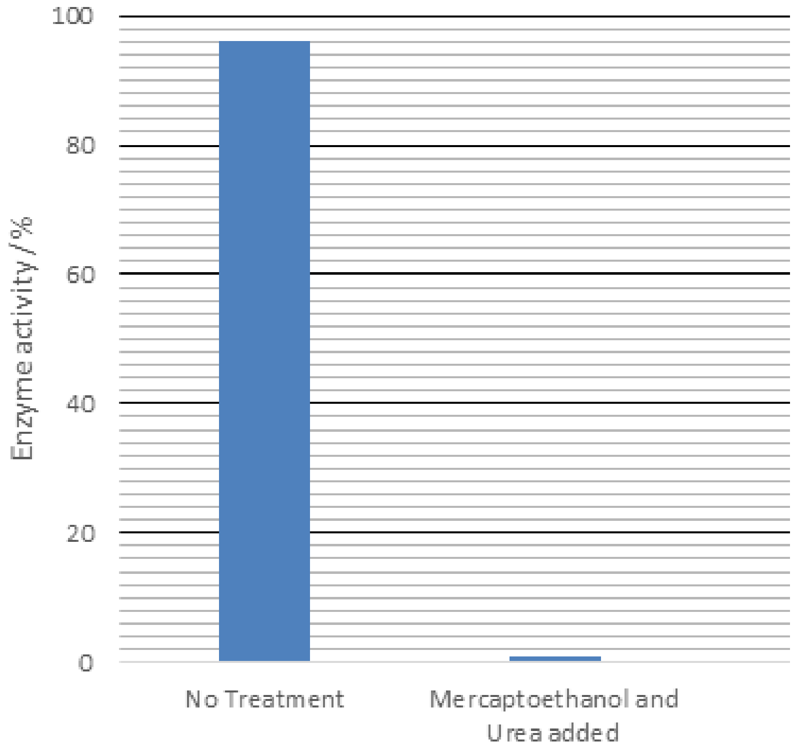

Fig. 1.2

(b) Using information from Fig. 1.1 and 1.2, explain the enzyme activity after mercaptoethanol and urea were added. [3]

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For Examiner's Use

Using the radioactive-labelled DNA, a cell with condensed chromosomes was carefully observed and the following measurements were taken:

For Examiner's Use

A – distance from pole to pole
B – distance between a centromere and the respective poles

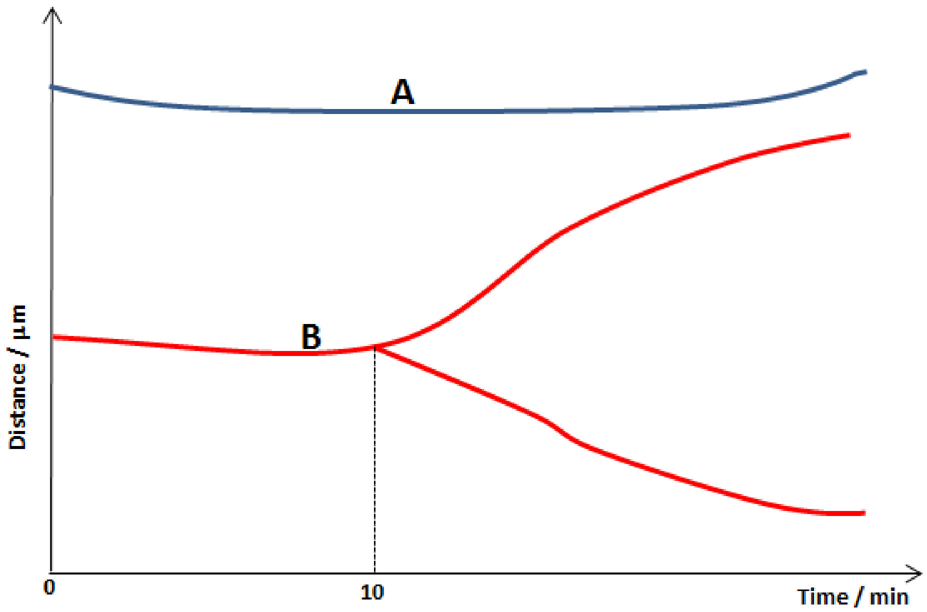

Fig. 1.3

(c) Describe the event that occurred in **B** after 10 minute. [3]

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(d) Some drugs which are used in the treatment of cancer affect the function of spindle fibres. Suggest what would happen if the drug is introduced into the cell at 0 min. [2]

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[Total: 10]

2 Fig. 2.1 shows a molecule of transfer RNA (tRNA) attached to a corresponding enzyme.

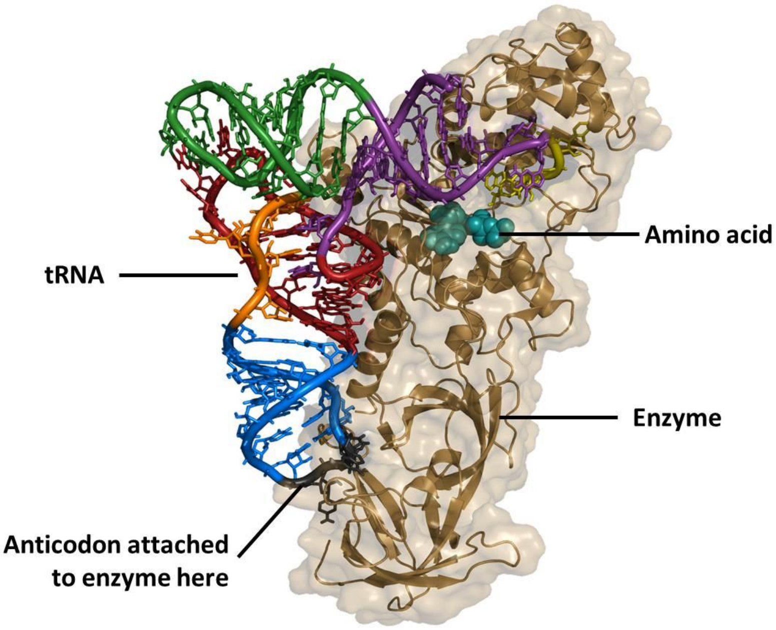

Fig. 2.1

(a) Identify the enzyme labelled in Fig. 2.1. [1]

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(b) Describe how the structure of the tRNA molecule and the enzyme molecule is maintained. [2]

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A specific example of the enzyme in Fig. 2.1 allows the amino acid histidine to bind to its active site. A single base substitution occurred in the gene coding for this enzyme results in the enzyme recognizing tyrosine instead of histidine.

For Examiner's Use

(c) Suggest why a mutation in this gene would have extremely damaging effects on the organism. [3]

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(d) The flow of genetic information from DNA to polypeptide is an accurate process. Explain how this is so. [4]

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[Total: 10]

3 Fig. 3.1 summarises the reactions which occur in the Calvin cycle.

For Examiner's Use

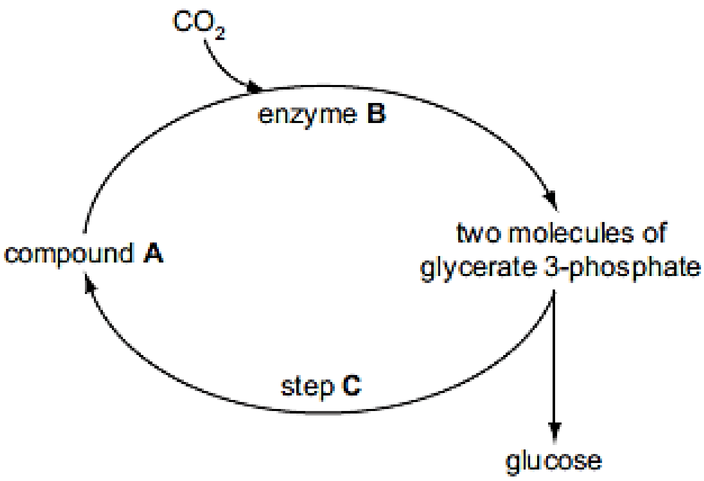

Fig. 3.1

(a) Name

(i) Compound **A** [1]

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(ii) Enzyme **B** [1]

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The Calvin cycle is part of the light-independent reactions of photosynthesis. These reactions continue when a plant is moved from light conditions to dark conditions, but only for a very short time.

(b) With reference to Fig. 3.1, explain why this is the case. [3]

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Fig. 3.2 outlines the early stages of respiration.

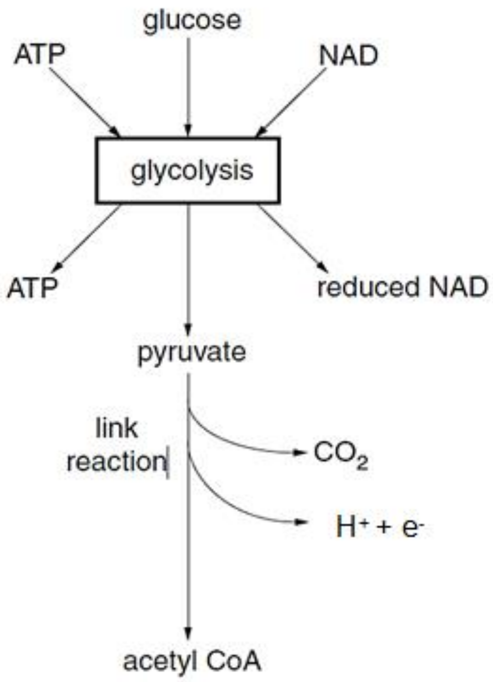

Fig. 3.2

With reference to Fig. 3.2,

(c) **(i)** explain why ATP is needed at the start of glycolysis. [1]

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(ii) explain the fate of the hydrogen atoms released by pyruvate during the link reaction. [3]

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(d) State why ATP is regarded as the universal energy currency in organisms. [2]

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[Total: 11]

4 A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 4.1.

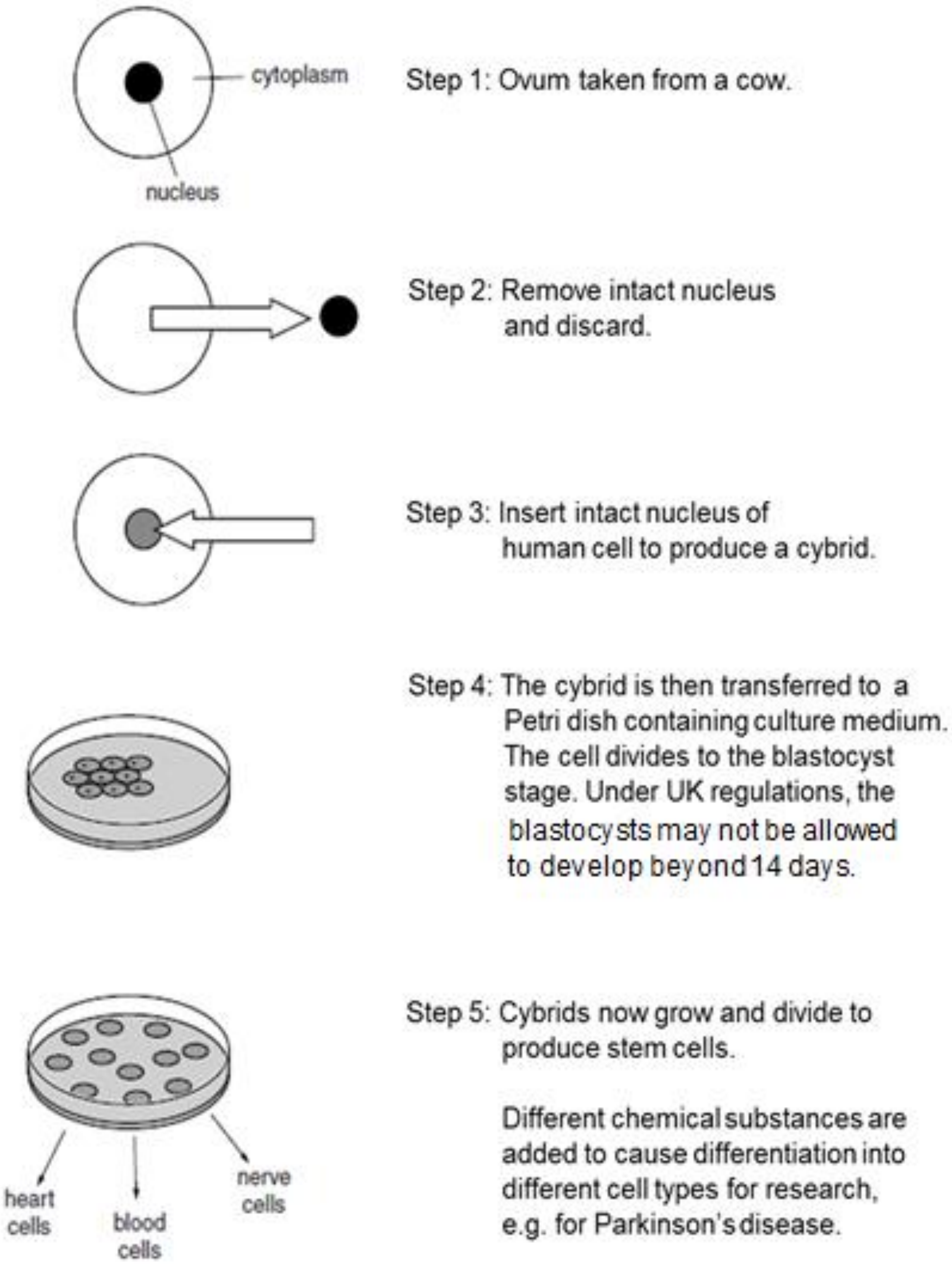

Fig. 4.1

For Examiner's Use

For Examiner's Use

(a) **(i)** Describe the unique features of the stem cells obtained in step 4. [2]

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(ii) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. State why there is DNA in the cytoplasm of the cybrid. [1]

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(iii) The development of cybrids is an important source of stem cells to be used in medical researches. However, some people argue that it is unethical to allow the production of cybrid.

Suggest why the production of cybrids may be considered unethical. [1]

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(b) A recombinant plasmid was produced by inserting a gene into a 19kb plasmid at *EcoRI* restriction site. The 19kb plasmid and the recombinant plasmid was then cleaved separately using a different restriction enzyme *HpnI*. The resultant fragments were then separated using gel electrophoresis. The results are shown below:

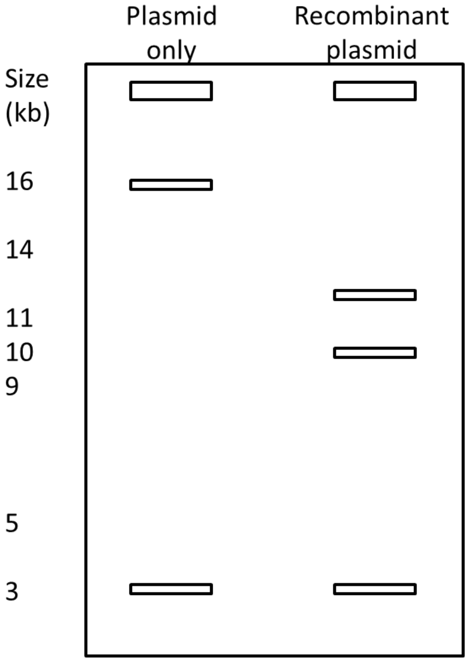

Fig. 4.2

For Examiner's Use

(i) Based on the information provided, draw and label the recombinant plasmid produced by the scientist. On your diagram, clearly show the positions of the restriction sites. [2]

(ii) Outline how the band pattern in Fig. 4.2 was obtained. [3]

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[Total: 9]

For Examiner's Use

Section B

Answer **ONE** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be in set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 5

(a) Describe the similarities and differences between homologous chromosomes. [6]

(b) Outline the process of the photolysis of water and describe what happens to the products of photolysis. [6]

(c) Discuss the implications of Genetically Modified Organisms. [8]

[Total: 20]

QUESTION 6

(a) Describe the similarities and differences between a secretory vesicle and a lysosome. [6]

(b) Outline how the frequency of sickle cell anaemia allele is maintained in malaria affected area. [8]

(c) Discuss the ethical concerns for the use of human genome project. [6]

[Total: 20]

End of Paper